Power module (hw/power-module) — one PCB, assembled twice by JLC: dash node and main node. docs/design-power-module.md.

Input and outputs

File: input.kicad_sch

3V3 buck

File: buck3v3.kicad_sch

5V buck

File: buck5v.kicad_sch

Input and outputs: J1 -> XF1/F1 (5 A dash, 2 A main) -> D1 SMCJ24CA -> Q1 AOD4185 -> D2 SMCJ24CA, C1 = 12V_PROT;
RT2 -> J3 (main feed / relay coils); Q2 (EN_A, default OFF) -> RT1 -> J2 (knob VM); 12V_SENSE divider -> J5.
3V3 buck: U1 TPS54360B always on (EN open) -> 3V3 (J5.1). 5V buck: U2 TPS54360B, EN_5V default OFF, 22 uF hold -> 5V0 (J4).
Main copy: J4, J2, RT1 not fitted; EN_5V and EN_A float -> U2 and Q2 stay off (R11, R13).
Owner-fitted through-hole parts are DNP for the JLC assembly: J1-J5, XF1, C1, RT1, RT2 (F1 is a loose blade, BOM only).
Footprints: KiCad stock libraries + project library PWRMOD (RXEF PTC, flagged VERIFY).

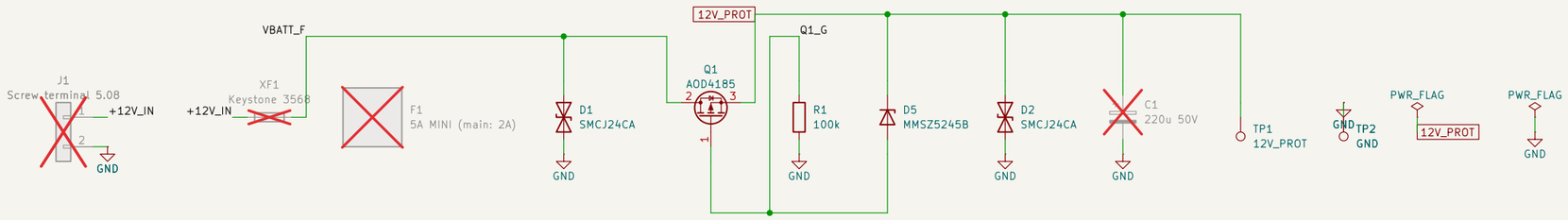
Embedded symbols (lib PWRMOD); generated by scripts/gen-power-module
Source: docs/design-power-module.md (2026-10-28, owner-approved)
Miata NA hardware — power module (JLC, built twice)

Sheet: /
File: power-module.kicad_sch

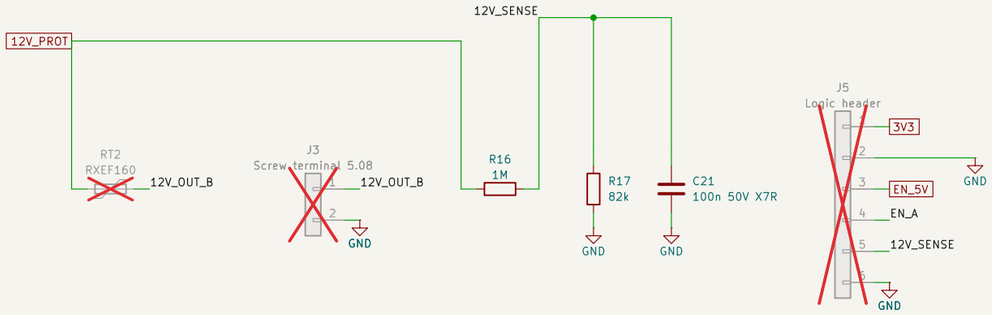
Title: Power module — dash + main node (root)

Size: A3	Date: 2026-10-28	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 1/4

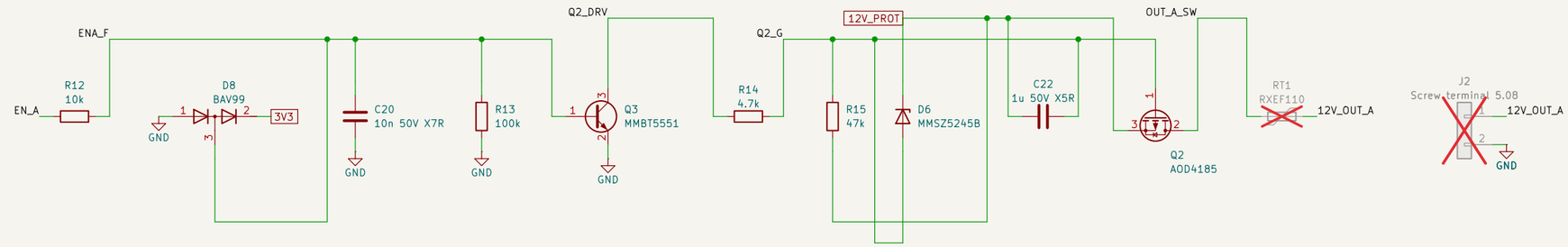
J1 -> XF1/F1 MINI 5 A (main copy 2 A) -> D1 SMCJ24CA -> Q1 AOD4185 reverse-polarity P-FET -> D2 SMCJ24CA, C1 = 12V_PROT



Main feed J3 (RT2 RXEF160, always on) and 12V_PROT sense (k = 82k/1082k = 0.0758, f_c 21 Hz) to J5



Knob VM switch: EN_A -> R12/D8/C20/R13 -> Q3 MMBT5551 -> Q2 AOD4185 high side (C22 slew, default OFF) -> RT1 RXEF110 -> J2



Key values (docs/design-power-module.md sections 4, 9-11)

F1 5 A: I_in(9 V) 2.729 A = 55 % cold / 64 % at +85 C. Main copy 2 A: 0.873 A = 44 % / 51 %
Q1: R_DS = 23.3 mOhm (V_GS -8.85 V, 110 C); P = 2.729^2 x 0.0233 = 0.174 W; T_J = 100 + 0.174 x 60 = 110 C
D5/D6 15 V zener: |V_GS| <= 15.75 V < 20 V; Pulse 1 with D1: V_DS(Q1) - -34.4 V < 40 V
12V_PROT: 8.85 V at 9 V in, <= 38.9 V in a load dump; -14 V reverse: 0 V
Q2 drive: V_GS = -0.909 x 12V_PROT until D6 clamps; Q3 base 2.64 V -> I_B 0.18 mA; R13 100k = default OFF
Sense: 38.9 V -> 2.95 V (3.00 V worst tol.) < 3.3 V; f_c = 1/(2 pi x 75.8k x 100n) = 21 Hz; 11.6 uA
RT2 RXEF160: 0.96 A hold at +85 C [E] >= 0.873 A; RT1 RXEF110: 0.66 A >= 0.5 A

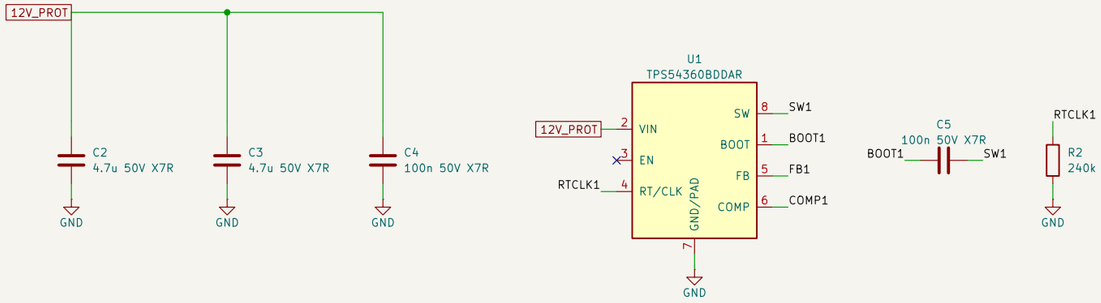
Embedded symbols (lib PWRMOD): generated by scripts/gen-power-module
Source: docs/design-power-module.md (2026-10-28, owner-approved)
Miata NA hardware — power module (JLC, built twice)

Sheet: /Input and outputs/
File: input.kicad_sch

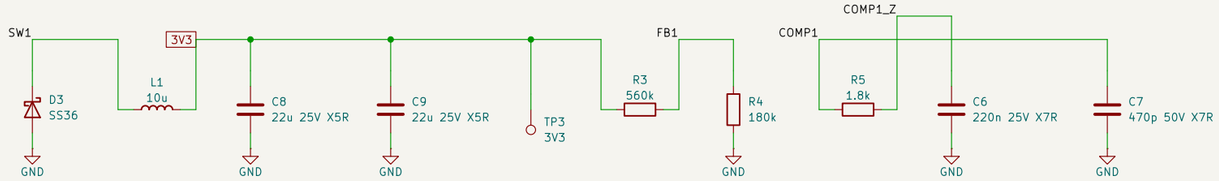
Title: Power module — input chain, 12 V outputs, sense, J5

Size: A3	Date: 2026-10-28	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 2/4

Input ceramics and U1 (EN open = enabled by the internal 1.2 uA pull-up)



Power stage, feedback (3.289 V) and compensation (f_co 9.8 kHz)



Key values (docs/design-power-module.md sections 5, 6)

f_sw = 92417 x 240^-0.991 = 404.5 kHz; t_SS = 1024/404.5 kHz = 2.5 ms
V_out = 0.8 x (1 + 560k/180k) = 3.289 V (3.22-3.36 V)
f_p = 0.3/(2 pi x 3.289 x 30u) = 486 Hz; f_co = 9.9 kHz; R5 1.8k, C6 220n (zero 402 Hz), C7 470p (pole 188 kHz)
Ripple (10 uH): 0.51 A at 8.85 V ... 0.74 A at 38.9 V
Sleep: 146 uA typ / 175 uA max non-switching; 270 uA no-load input (TI example)

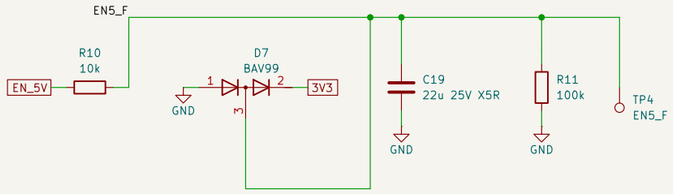
Embedded symbols (lib PWRMOD); generated by scripts/gen-power-module
Source: docs/design-power-module.md (2026-10-28, owner-approved)
Miata NA hardware — power module (JLC, built twice)

Sheet: /3V3 buck/
File: buck3v3.kicad_sch

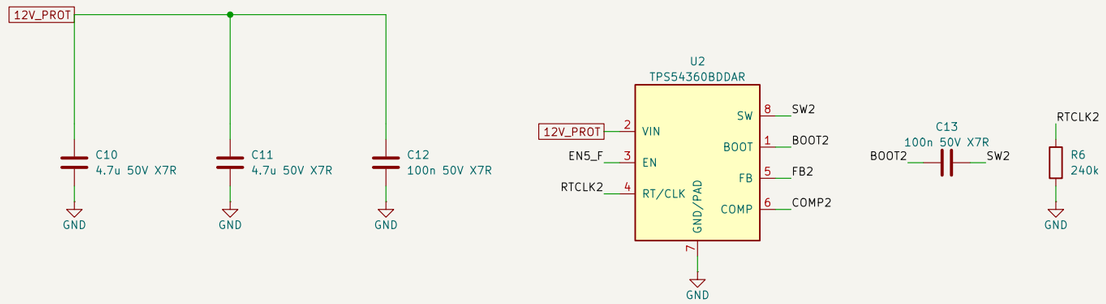
Title: Power module — U1 TPS54360B 3V3 always-on

Size: A3	Date: 2026-10-28	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 3/4

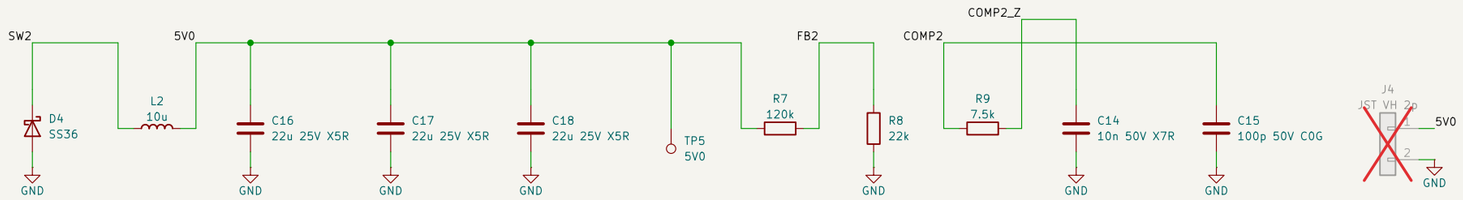
EN_5V: R10 10k + D7 clamp + C19 22u hold + R11 100k pull-down (default OFF, holds >= 0.92 s through an ESP32 reset)



Input ceramics and U2



Power stage, feedback (5.164 V), compensation (f_co 19.8 kHz), J4 to the Pi



Key values (docs/design-power-module.md sections 7, 8)

V_out = 0.8 x (1 + 120k/22k) = 5.164 V (5.06-5.27 V): Pi sees ~5.06 V at 2.3 A
f_p = 2.3/(2 pi x 5.164 x 39.3u) = 1.80 kHz; f_co 19.8 kHz; R9 7.5k, C14 10n (zero 2.12 kHz), C15 100p
EN: OFF 1.8 uA x 100k = 0.18 V < 1.1 V; ON 2.64 x 100/110 = 2.40 V > 1.3 V; hold 1.5 s x ln(2.40/1.3) = 0.92 s min
U2 P_IC (9 V, R_DS max) 0.70 W -> T_J = 85 + 15 + 0.70 x 55 = 138 C < 150 C
Re-crank: Pi >= 4.75 V while V_BATT >= 5.71 V; full regulation >= 6.03 V. No 5 V PTC (X10)

Embedded symbols (lib PWRMOD): generated by scripts/gen-power-module
Source: docs/design-power-module.md (2026-10-28, owner-approved)
Miata NA hardware - power module (JLC, built twice)

Sheet: /5V buck/
File: buck5v.kicad_sch

Title: Power module - U2 TPS54360B 5V0 (Pi), EN default OFF

Size: A3	Date: 2026-10-28	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 4/4